

Sandstorms and Duststorms

In arid and semiarid areas, high winds can whip up sand and dust to form powerful storms, blocking out the Sun and stripping away soil

A severe sand- or duststorm can create darkness in the middle of the day and reduce visibility to virtually nothing. These storms are formed when strong winds and dry conditions cause surface materials—dust, soil, or sand—to be lifted in a process called saltation. The wind carries particles a short distance before they fall back to the ground, dislodging other particles and accelerating the process. They remain suspended in the lower atmosphere.

The height and extent of the storm depends on the strength and persistence of the wind and the size of the particles. Wind gusts often exceed 50 mph (80 kph). Dust particles are smaller and are carried higher than sand—exceptionally to 20,000 ft (6,100 m). Some storms are driven by cold fronts (the leading

edge of a mass of cold air pushing under warmer air) while others, called haboobs (see panel, below), are created by strong downdrafts from storm cells.

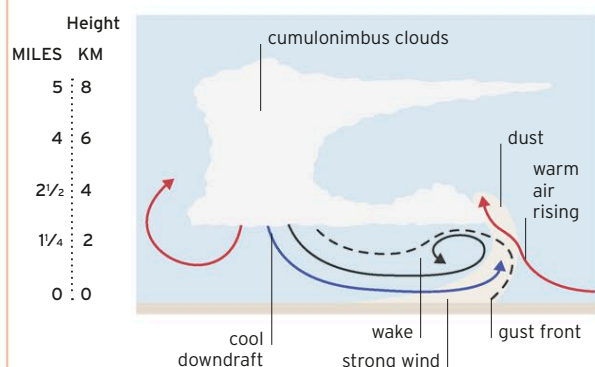
Cold fronts

The most notorious duststorms struck the High Plains of the US on April 14, 1935, turning a warm, sunny afternoon into a choking, zero-visibility blackout in parts of Oklahoma, Texas, Kansas, Nebraska, and Missouri. A combination of an extended drought, poor soil management, and high winds produced by a rapidly moving cold front created several huge duststorms. The great Australian duststorm of 2009, also created by a cold front, carried 2.75 million tons (2.5 million tonnes) of the Australian continent into the ocean.



HOW A HABOOB FORMS

Derived from the Arabic word *habb*, meaning “wind,” a haboob is formed when strong winds blow out from a storm cell. The cell forms when warm, moist air rises, but air within that updraft is cooled by evaporating rain. This air may form a powerful downdraft and outflow, led by a gust front. As the front hits dry ground in advance of the storm, it kicks up surface dust or sand. The gust front produces a “wall” of dust, which may spread over a larger area than the storm itself.



▽ RED DUST

Sydney Harbor Bridge was shrouded in red dust during the great duststorm of September 2009. This is recorded to have been the worst to hit eastern Australia in the last 70 years.

▷ A CITY ENGULFED

A billowing wall of dust can be seen as it is about to engulf downtown Phoenix, Arizona, in July 2011. The dust cloud was generated by a haboob, one of several to affect the city that summer.

